Measure Theoretic Probability
Lesson 22
Subsequences and Convergence; Expectation of Simple Random Variables

Kirschner Awonatey Amaanya

Department of Statistics and Actuarial Science, KNUST

Outline

1. Recap: Subsequences of Real Numbers
2. Subsequences and Convergence in Probability
3. A Subsequence Converging Almost Surely
4. A Characterisation of Convergence in Probability
5. Expectation of Simple Random Variables
6. Summary and Exercises

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Subsequences of Convergent Sequences

Fact 1

If $a_n \to a$ in $\mathbb{R}$, then every subsequence $a_{n_k} \to a$ as $k \to \infty$.

Idea

The whole sequence is eventually trapped in any small neighbourhood of a. A subsequence only selects some of its terms, so it is trapped there too.

Fact 2: a divergent sequence may still have convergent subsequences

$0, 1, 0, 1, \ldots$ does not converge. The odd-indexed terms give $0, 0, 0, \cdots \to 0$ and the even-indexed terms give $1, 1, 1, \cdots \to 1$.

Fact 3: but not always

$1, 2, 3, 4, \ldots$ has **no** convergent subsequence: every subsequence still increases without bound.

Bolzano–Weierstrass

Theorem (Bolzano–Weierstrass)

Every **bounded** sequence of real numbers has a convergent subsequence.

Idea of the proof: bisection

The terms lie in a horizontal strip. Cut the strip in half: at least one half contains infinitely many terms, so keep it. Cut it in half again and keep a half with infinitely many terms, and so on. The strips get arbitrarily narrow, and the chosen terms squeeze down to a limit.

Our Goal here

We prove random-variable analogues for $\xrightarrow{\mathbb{P}}$ and $\xrightarrow{\text{a.s.}}$. Random variables are not numbers, so the proofs bring in numbers through probabilities.

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Theorem 1: Subsequences Inherit $\xrightarrow{\mathbb{P}}$

Theorem 1

If $X_n \xrightarrow{\mathbb{P}} X$, then every subsequence $X_{n_k} \xrightarrow{\mathbb{P}} X$ as $k \to \infty$.

Convention on indices

A subsequence has indices $n_1 < n_2 < n_3 < \cdots$. Consequently

$$n_k \geq k \qquad \text{for all } k.$$

(If nothing is skipped, $n_k = k$. Once a term is skipped, $n_k > k$ from then on.) We use this convention repeatedly.

Proof of Theorem 1

Setup

Fix $\varepsilon > 0$. We must show $\mathbb{P}(|X_{n_k} - X| > \varepsilon) \to 0$ as $k \to \infty$. Let $\delta > 0$ be arbitrary.

Use convergence of the full sequence

Since $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$, there is N such that

$$\mathbb{P}\big(|X_n - X| > \varepsilon\big) < \delta \qquad \text{for all } n \geq N.$$

Pass to the subsequence

If $k \geq N$, then $n_k \geq k \geq N$, so

$$\mathbb{P}\big(|X_{n_k} - X| > \varepsilon\big) < \delta \qquad \text{for all } k \geq N.$$

As $\delta > 0$ was arbitrary, $\lim_{k\to\infty} \mathbb{P}(|X_{n_k} - X| > \varepsilon) = 0$. Hence $X_{n_k} \xrightarrow{\mathbb{P}} X$. □

Almost sure version

If $X_n \xrightarrow{\text{a.s.}} X$, then every subsequence $X_{n_k} \xrightarrow{\text{a.s.}} X$. The proof is similar (it is a statement about the real sequences $X_n(\omega)$, ω in the convergence set.

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Theorem 2

Theorem 2

If $X_n \xrightarrow{\mathbb{P}} X$, then there exists a subsequence X_{n_k} with

$$X_{n_k} \xrightarrow{\text{a.s.}} X \qquad (k \to \infty).$$

Why it is interesting

Starting from the **weaker** mode of convergence, we find inside the sequence a subsequence that converges in the **stronger** mode.

Tool: sufficient condition (4) from Lesson 21

If $\sum_{k=1}^{\infty} \mathbb{P}\big(|Y_k - X| > \frac{1}{k}\big) < \infty$, then $Y_k \xrightarrow{\text{a.s.}} X$.

Proof of Theorem 2

Step 1: choose the indices

Since $X_n \xrightarrow{\mathbb{P}} X$, for each $k = 1, 2, 3, \ldots$ we have $\mathbb{P}\big(|X_n - X| > \frac{1}{k}\big) \to 0$ as $n \to \infty$. So we can pick n_k with

$$\mathbb{P}\Big(|X_{n_k} - X| > \tfrac{1}{k}\Big) < \frac{1}{2^k}.$$

Choose them inductively with $n_k > n_{k-1}$: if the first choice is too small, look further along the sequence, since the probabilities stay small beyond it.

Step 2: sum the probabilities

$$\sum_{k=1}^{\infty} \mathbb{P}\Big(|X_{n_k} - X| > \tfrac{1}{k}\Big) \leq \sum_{k=1}^{\infty} \frac{1}{2^k} = 1 < \infty.$$

Step 3: conclude

Put $Y_k = X_{n_k}$. Then $\sum_k \mathbb{P}(|Y_k - X| > 1/k) < \infty$, so by the sufficient condition $Y_k \xrightarrow{\text{a.s.}} X$. That is, $X_{n_k} \xrightarrow{\text{a.s.}} X$. □

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Theorem 3

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$X_n \xrightarrow{\mathbb{P}} X$ if and only if **every** subsequence X_{n_k} has a further subsequence $X_{n_{k_j}}$ with $X_{n_{k_j}} \xrightarrow{\text{a.s.}} X$ as $j \to \infty$.

Proof of ($\Rightarrow$)

Assume $X_n \xrightarrow{\mathbb{P}} X$ and let X_{n_k} be any subsequence. By Theorem 1, $X_{n_k} \xrightarrow{\mathbb{P}} X$. Apply Theorem 2 to the sequence $(X_{n_k})_k$: it has a subsequence converging almost surely to X. □

Use

Statements like this are rarely interesting on their own, but they are the tool we use later to prove results we do care about.

Proof of ($\Leftarrow$): by Contradiction

Assume

Every subsequence has a further subsequence converging a.s. to X. Suppose, for contradiction, that $X_n \xrightarrow{\mathbb{P}} X$.

Consequence of the negation

There is $\varepsilon > 0$ with $\mathbb{P}(|X_n - X| > \varepsilon) \not\to 0$. So there are $\delta > 0$ and **infinitely many** n with

$$\mathbb{P}\big(|X_n - X| > \varepsilon\big) > \delta.$$

List these indices increasingly as $n_1 < n_2 < \cdots$. This gives a subsequence X_{n_k} with $\mathbb{P}(|X_{n_k} - X| > \varepsilon) > \delta$ for **all** k.

Contradiction

By hypothesis, X_{n_k} has a further subsequence $X_{n_{k_j}} \xrightarrow{\text{a.s.}} X$. Almost sure convergence implies convergence in probability, so $\mathbb{P}(|X_{n_{k_j}} - X| > \varepsilon) \to 0$ as $j \to \infty$. But all these probabilities exceed $\delta > 0$.

Contradiction. Hence $X_n \xrightarrow{\mathbb{P}} X$. □

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Expectation: the Discrete Definition

Reminder

For a discrete random variable X,

$$\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X = x),$$

summed over all possible values x (not necessarily integers; values that X cannot take contribute 0).

Today's setting

We now look at this from the measure-theoretic point of view, for **simple** random variables only. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

Purpose

Expectation gives us a third way to bring real numbers into the study of convergence of random variables.

Definition for Simple Random Variables

Simple random variable

$X : \Omega \to \mathbb{R}$ is simple if

$$X(\omega) = \sum_{i=1}^{n} x_i \, I_{A_i}(\omega), \qquad x_i \in \mathbb{R}, \quad A_i \in \mathcal{F}.$$

Expectation

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \, \mathbb{P}(A_i).$$

The sets A_i must lie in $\mathcal{F}$, so that $\mathbb{P}(A_i)$ makes sense.

Question

The same X may have several representations of this form. Is $\mathbb{E}[X]$ independent of the representation?

The Canonical Partition

Definition (terminology of the lectures)

Write $X = \sum_{i=1}^{n} x_i I_{A_i}$ with

- the values $x_1, \ldots, x_n$ **distinct**, and
- $A_i = X^{-1}(\{x_i\}) = \{X = x_i\}$, so that $A_1, \ldots, A_n$ partition Ω.

If two sets mapped to the same number we would merge them, so we may always assume this form.

Agreement with the discrete formula

Since $\mathbb{P}(A_i) = \mathbb{P}(X = x_i)$,

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i \, \mathbb{P}(A_i) = \sum_{i=1}^{n} x_i \, \mathbb{P}(X = x_i),$$

which is the usual definition. Nothing new is happening.

Remark

The A_i need not be intervals or even connected. Each A_i may consist of several pieces of Ω.

Two Representations Force Consistency

Setup

Let $X = \sum_{i=1}^{n} x_i I_{A_i} = \sum_{j=1}^{m} y_j I_{B_j}$, where $\{A_i\}$ is the canonical partition and $\{B_j\}$ is another partition of Ω into sets of $\mathcal{F}$.

Key observation

Suppose $\omega \in A_i \cap B_j$. Evaluating $X(\omega)$ in the two representations gives $x_i = X(\omega) = y_j$. Hence

$$x_i = y_j \quad \text{whenever} \quad A_i \cap B_j \neq \varnothing.$$

Example: why not an arbitrary partition

Suppose B_1 meets both A_1 and A_2. Then $x_1 = y_1$ and $x_2 = y_1$, so $x_1 = x_2$. This contradicts the distinctness of the x_i in the canonical partition. So a second partition must be **finer** than the canonical one (each B_j lies inside a single A_i).

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Summary

Key results

Concept / Result	Content
Real subsequences	$a_n \to a \Rightarrow a_{n_k} \to a$; bounded $\Rightarrow$ a convergent subsequence exists
Index convention	$n_1 < n_2 < \cdots$, hence $n_k \geq k$
Theorem 1	$X_n \xrightarrow{\mathbb{P}} X \Rightarrow X_{n_k} \xrightarrow{\mathbb{P}} X$ (and similarly for $\xrightarrow{\text{a.s.}}$)
Theorem 2	$X_n \xrightarrow{\mathbb{P}} X \Rightarrow \exists\, X_{n_k} \xrightarrow{\text{a.s.}} X$ (choose $\mathbb{P}(\lvert X_{n_k} - X\rvert > \frac{1}{k}) < 2^{-k}$)
Theorem 3	$X_n \xrightarrow{\mathbb{P}} X \iff$ every subsequence has a further subsequence $\xrightarrow{\text{a.s.}} X$
Simple r.v.	$X = \sum x_i I_{A_i}$, $\mathbb{E}[X] = \sum x_i \mathbb{P}(A_i)$
Canonical partition	x_i distinct, $A_i = \{X = x_i\}$

Exercises

1. Prove Fact 1: if $a_n \to a$, then $a_{n_k} \to a$. Use $n_k \geq k$.
2. Prove that if $X_n \xrightarrow{\text{a.s.}} X$, then every subsequence $X_{n_k} \xrightarrow{\text{a.s.}} X$.
3. In the moving-indicator example of Lesson 21, find a subsequence of $X_n = \omega + I_{[a_n, b_n]}$ that converges almost surely to $X(\omega) = \omega$.
4. In the proof of Theorem 2, explain how to choose $n_k > n_{k-1}$ at each step.
5. Show that if $X_n \xrightarrow{\mathbb{P}} X$, then $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for the subsequence along the even indices, without using Theorem 1.
6. Let X take values $1, 2, 3$ on a partition A_1, A_2, A_3 with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$. Compute $\mathbb{E}[X]$.
7. Let $X = 2I_A + 2I_B$ with $A, B \in \mathcal{F}$ disjoint. Write X in canonical form and compute $\mathbb{E}[X]$.
8. In the contradiction proof of Theorem 3, why do infinitely many n satisfy $\mathbb{P}(|X_n - X| > \varepsilon) > \delta$ for some fixed $\delta > 0$?

Outlook

Next lesson

We prove that the expectation of a simple random variable does not depend on the representation chosen, and then develop its properties.

Looking ahead

With expectation available, we can study a further type of convergence, convergence of expected values, and compare it with $\xrightarrow{\text{a.s.}}$ and $\xrightarrow{\mathbb{P}}$.